Qatar-2b

R-0926

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-2_180525` · created 2026-07-29T10:32:55+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458263.73627 +/- 0.00063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.227 +/- 0.062
Transit depth (Rp/Rs)^2	5.16 +/- 2.8 [%]
Orbital Inclination (inc)	88.95 +/- 1.75
Airmass coefficient 1 (a1)	11.85 +/- 1.58
Airmass coefficient 2 (a2)	-1.77 +/- 0.23
Scatter in the residuals of the lightcurve fit is	11.74 %
Best Comparison Star	#2 - [54, 248]
Optimal Aperture	10.17
Optimal Annulus	14.4
Transit Duration (day)	0.0799 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.227 ± 0.062 vs 0.16327 (deviation 1.03σ)
Duration fit vs published	0.0799 d vs 0.0767 d (deviation 0.67σ)
Mid-time O–C	1.6 min
Depth SNR	3.7
Residual scatter	11.74 %

Light curve

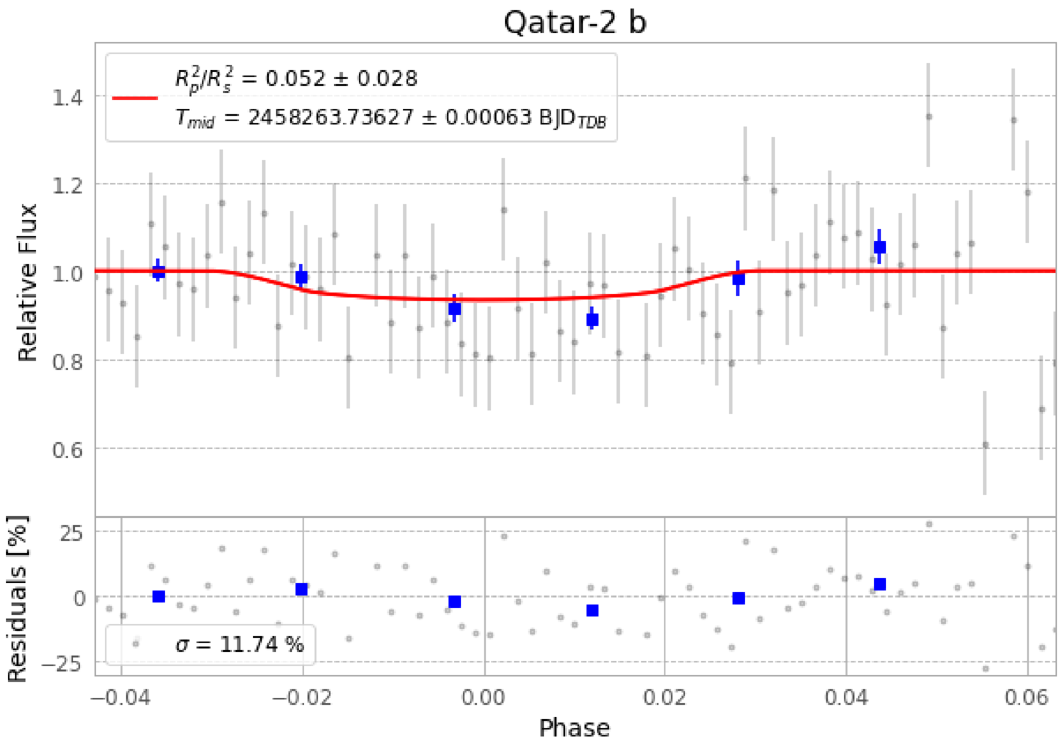

Figure 1 — FinalLightCurve_Qatar-2 b_2018-05-24.png (SHA-256 bb3c84e804296c7f…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [409, 436], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 4f640d0d7c5953f3…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

71 FITS frames (47 MB), `Qatar-2180525040912.FITS` → `Qatar-2180525073912.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-180525.log (c8d494250285…), FinalLightCurve_Qatar-2 b_2018-05-24.png (bb3c84e80429…), FinalLightCurve_Qatar-2 b_2018-05-24.csv (3d63e9784fb0…)

Compute resources

Wall time 350.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)